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Deep Learning And Applied Computer Vision

ASSIGNMENT 2

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1. Dataset

1.1 Cityscapes Dataset Overview

The Cityscapes Dataset focuses on semantic understanding of urban street scenes. In the following, we give an overview on the design choices that were made to target the dataset's focus.

1.2 Features - Cityscapes Dataset

Features	Descriptions
Polygonal Annotations	● Dense semantic segmentation ● Instance segmentation for vehicle and people
Complexity	● 30 classes
Diversity	● 50 cities ● Several months (spring, summer, fall) ● Daytime ● Good/medium weather conditions ● Manually selected frames ○ Large number of dynamic objects ○ Varying scene layout ○ Varying background
Volume	● 5 000 annotated images with fine annotations ● 20 000 annotated images with coarse annotations
Metadata	● Preceding and trailing video frames. Each annotated image is the 20th image from a 30 frame video snippets (1.8s) ● Corresponding right stereo views ● GPS coordinates ● Ego-motion data from vehicle odometry ● Outside temperature from vehicle sensor

1.3 Class Definitions - Cityscapes Dataset

2. Assignment 2

2.1 Data split

Subset	Number of Images	Purpose
Training	40,000	Model training
Validation	10,000	Hyperparameter tuning and model selection
Testing	10,000	Final evaluation

2.2 Data augmentation

The image processing pipeline consists of two distinct workflows for training and evaluation:

- Train Split: Raw Image -> Random Crop -> Normalization -> Model Input
- Test/Val Split: Raw Image -> Normalization -> Model Input

Using PyTorch's torchvision.transforms, the data augmentation pipeline for the training set consists of the following transformations:

Random Crop (transforms.RandomCrop)

- Parameters: size = 32, padding = 4, padding_mode = 'reflect'
- Mechanism: Expands the input 32x32 image to 40x40 via reflection padding, followed by a random 32x32 crop.
- Purpose: Introduces spatial translation invariance and prevents positional memorization.

Normalization (transforms.Normalize)

- Parameters:
  - Channel Mean: [0.5071, 0.4867, 0.4408]
  - Channel Std: [0.2675, 0.2565, 0.2761]
- Formula: X_norm = (X - mean) / std
- Purpose: Centers channel-wise distributions around zero mean and unit variance to stabilize backpropagation and prevent activation saturation in ViT block layers.

2.3 Chosen Topic

2.3.1 Overview

3. References

Asad, Muhammad. "Introduction to Softmax Classifier in PyTorch - MachineLearningMastery.com." *Machine Learning Mastery*, https://machinelearningmastery.com/introduction-to-softmax-classifier-in-pytorch/. Accessed 15 October 2026.